

INTROSPECTABLE, UPDATABLE, and UNCERTAINTY-AWARE Classification of Language Model INSTRUCTION-FOLLOWING



Reexpress AI

<https://re.express/>

Challenge

Existing closed-weight language models (LMs) are unreliable: Uninterpretable, overconfident, and difficult to calibrate.

Agent-based pipelines inherit these key limitations.

Inaccessibility of the hidden states complicates reliably estimating the predictive uncertainty.

Solution

Ensemble multiple closed-weight LMs and a representation model as input to a **SIMILARITY-DISTANCE-MAGNITUDE (SDM)** estimator.

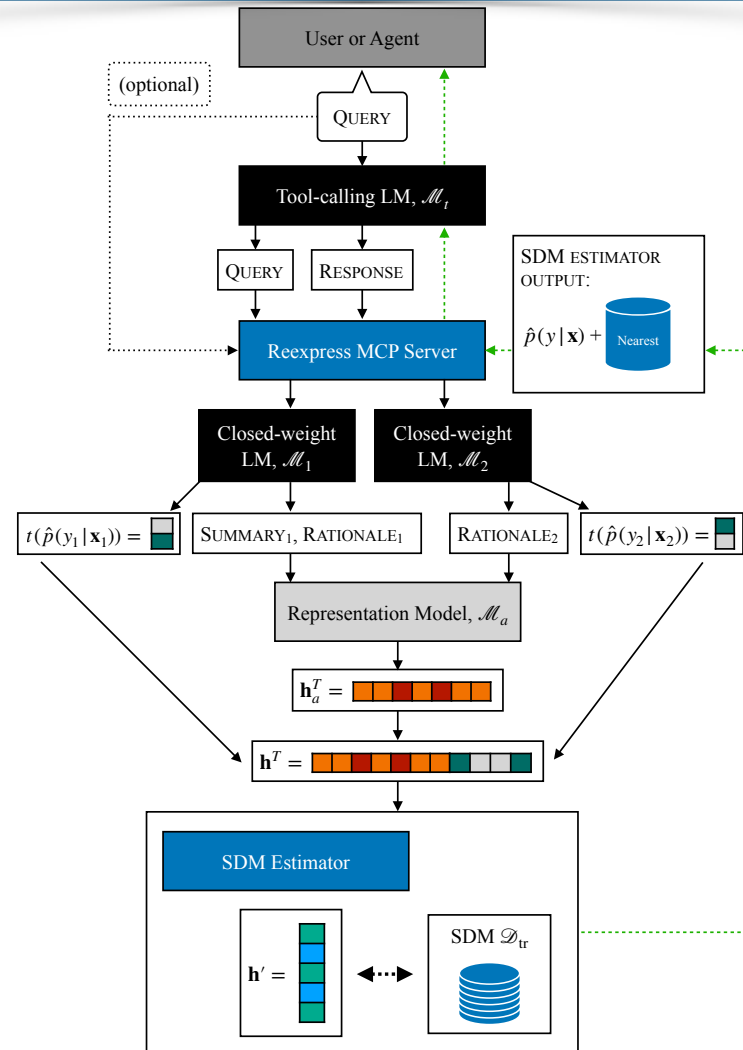
We release the **REEXPRESS MCP SERVER** and **open-source data (OPENVERIFICATION1)** to bootstrap domain-specific estimators for agentic pipelines, adding interpretability-by-exemplar, local *updatability* (calibrating the predictive uncertainty to modify test-time search graphs, without altering LM weights), and robust uncertainty quantification.

Task: Binary Classification of Instruction-following

Classify whether the generated response (RESPONSE) from $\mathcal{M}_t$ (a tool-calling LM) correctly and faithfully answers the user's instructions (QUERY), where the ground-truth labels correspond to $y \in \{0 = \text{Not Verified}, 1 = \text{Verified}\}$.

Data: Labeled training set, $\mathcal{D}_{tr} = \{(\text{Query}_n, \text{Response}_n, y_n)\}_{n=1}^N$ and a corresponding held-out calibration set, $\mathcal{D}_{ca}$

Seek: Prediction, $\hat{y}$, *and* an estimate of $p(y | \mathbf{x})$, the predictive uncertainty of that classification decision.

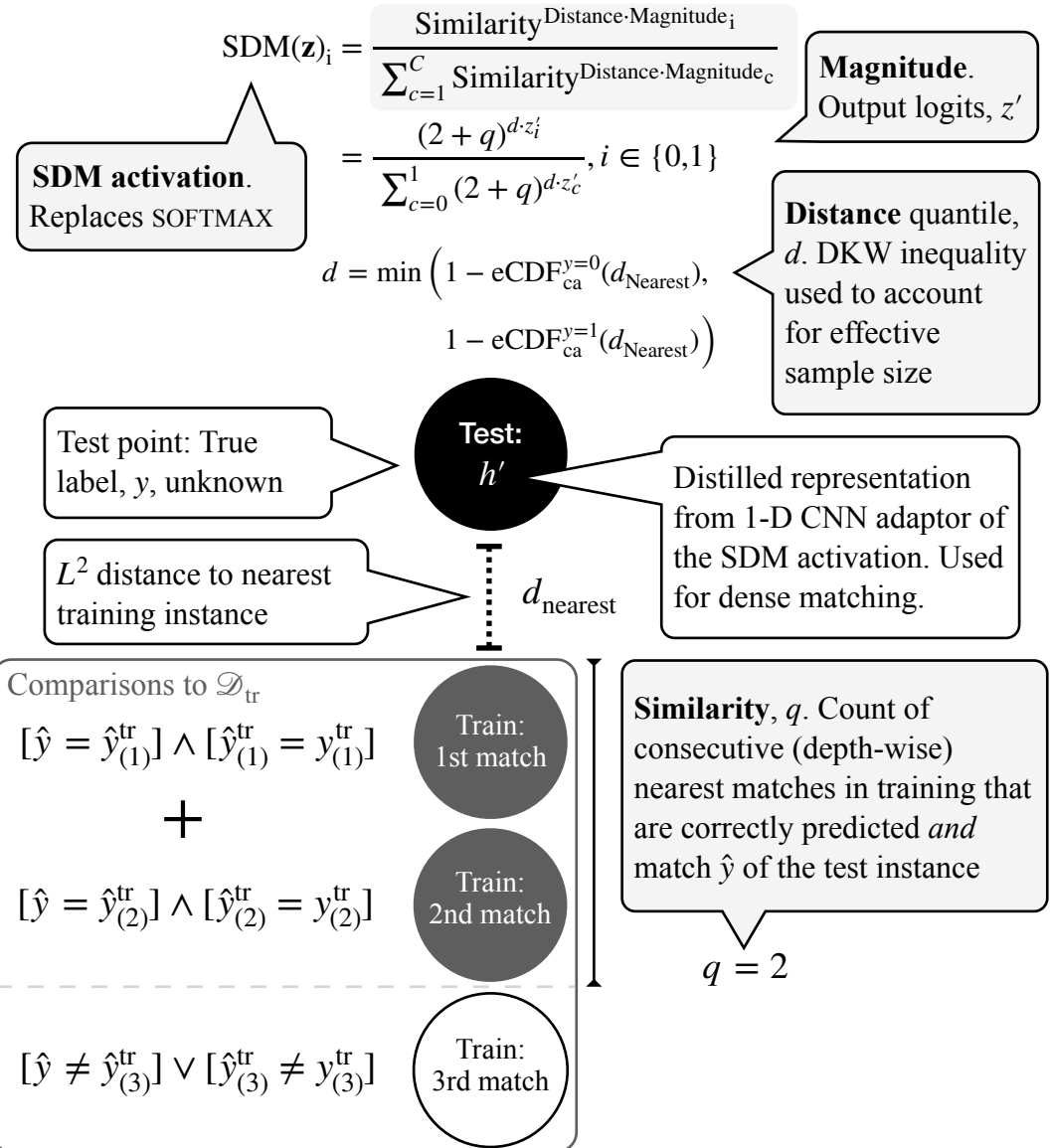


Actionable Interpretability. The REEXPRESS MCP SERVER provides $\hat{p}(y | \mathbf{x})$, a robust estimate of instruction-following of a closed-weight, tool-calling LM, $\mathcal{M}_t$, and interpretability-by-exemplar against a database of labeled examples (*introspection*), via an SDM estimator.

$\alpha = 0.9$		$y = 0$		$y = 1$		$y \in \{0,1\}$	
Dataset	Method	Acc.	$\frac{n}{ \mathcal{D}_{te} }$	Acc.	$\frac{n}{ \mathcal{D}_{te} }$	Acc.	$\frac{n}{ \mathcal{D}_{te} }$
HLE-TEST	GPT5.5+GEMINI3.1 VERBALIZED PROBABILITY	0.05	0.19	0.98	0.39	0.68	0.57
OPENTHOUGHTS-TEST	GPT5.5+GEMINI3.1 VERBALIZED PROBABILITY	0.95	0.48	0.88	0.44	0.92	0.92
MATHNET-TEST	GPT5.5+GEMINI3.1 VERBALIZED PROBABILITY	0.77	0.39	0.92	0.45	0.85	0.84
HLE-TEST	REEXPRESSV2.4.0 SDM ESTIMATOR (LOWER)	0*	<0.01	1	0.01	0.83*	0.01
OPENTHOUGHTS-TEST	REEXPRESSV2.4.0 SDM ESTIMATOR (LOWER)	0.96	0.34	0.98	0.30	0.97	0.64
MATHNET-TEST	REEXPRESSV2.4.0 SDM ESTIMATOR (LOWER)	0.96	0.21	0.94	0.17	0.95	0.38

*The one incorrect prediction corresponds to a ground-truth label annotation error.

Uncertainty-awareness: LMs as hidden instance-based metric learners



SDM estimator. Data-driven partitioning of the calibration set to estimate a region with class- and prediction-conditional accuracy at least α , a value near 1.

